MATH110: Confidence Intervals

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Learning Objectives

- Identify the critical value z* for a confidence level
- Compute the standard error and margin of error
- Construct a confidence interval for a population mean

Simplify each expression completely. Show all steps and circle your final answer.

Confidence interval for a mean (σ known)

1. A sample of n = 16 has a sample mean of 131. The population standard deviation is sigma = 10. Construct the 99% confidence interval for the population mean mu.

$$\bar{x} = 131, \quad \sigma = 10, \quad n = 16, \quad 99\%\ \text{CI}$$

Answer: ______________________________

2. A sample of n = 100 has a sample mean of 84. The population standard deviation is sigma = 20. Construct the 95% confidence interval for the population mean mu.

$$\bar{x} = 84, \quad \sigma = 20, \quad n = 100, \quad 95\%\ \text{CI}$$

Answer: ______________________________

3. A sample of n = 25 has a sample mean of 120. The population standard deviation is sigma = 15. Construct the 99% confidence interval for the population mean mu.

$$\bar{x} = 120, \quad \sigma = 15, \quad n = 25, \quad 99\%\ \text{CI}$$

Answer: ______________________________

4. A sample of n = 25 has a sample mean of 81. The population standard deviation is sigma = 12. Construct the 90% confidence interval for the population mean mu.

$$\bar{x} = 81, \quad \sigma = 12, \quad n = 25, \quad 90\%\ \text{CI}$$

Answer: ______________________________

5. A sample of n = 36 has a sample mean of 67. The population standard deviation is sigma = 20. Construct the 95% confidence interval for the population mean mu.

$$\bar{x} = 67, \quad \sigma = 20, \quad n = 36, \quad 95\%\ \text{CI}$$

Answer: ______________________________

6. A sample of n = 16 has a sample mean of 55. The population standard deviation is sigma = 15. Construct the 90% confidence interval for the population mean mu.

$$\bar{x} = 55, \quad \sigma = 15, \quad n = 16, \quad 90\% \text{ CI}$$

Answer: ____

7. A sample of n = 25 has a sample mean of 89. The population standard deviation is sigma = 15. Construct the 95% confidence interval for the population mean mu.

$$\bar{x} = 89, \quad \sigma = 15, \quad n = 25, \quad 95\% \text{ CI}$$

Answer: ____

8. A sample of n = 36 has a sample mean of 106. The population standard deviation is sigma = 20. Construct the 95% confidence interval for the population mean mu.

$$\bar{x} = 106, \quad \sigma = 20, \quad n = 36, \quad 95\% \text{ CI}$$

Answer: ____

9. A sample of n = 100 has a sample mean of 59. The population standard deviation is sigma = 15. Construct the 90% confidence interval for the population mean mu.

$$\bar{x} = 59, \quad \sigma = 15, \quad n = 100, \quad 90\% \text{ CI}$$

Answer: ____

10. A sample of n = 16 has a sample mean of 137. The population standard deviation is sigma = 10. Construct the 90% confidence interval for the population mean mu.

$$\bar{x} = 137, \quad \sigma = 10, \quad n = 16, \quad 90\% \text{ CI}$$

Answer: ____

Confidence Interval for a Population Proportion

11. In a random sample of 246 people, 39 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 39, \quad n = 246, \quad 99\% \text{ CI}$$

Answer: ____

12. In a random sample of 304 people, 49 had a particular trait. Construct a 90% confidence interval for the true proportion.

$$x = 49, \quad n = 304, \quad 90\% \text{ CI}$$

Answer: ____

13. In a random sample of 276 people, 47 had a particular trait. Construct a 95% confidence interval for the true proportion.

$$x = 47, \quad n = 276, \quad 95\% \text{ CI}$$

Answer: ________________________________

14. In a random sample of 127 people, 98 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 98, \quad n = 127, \quad 99\% \text{ CI}$$

Answer: ________________________________

15. In a random sample of 200 people, 53 had a particular trait. Construct a 90% confidence interval for the true proportion.

$$x = 53, \quad n = 200, \quad 90\% \text{ CI}$$

Answer: ________________________________

16. In a random sample of 200 people, 178 had a particular trait. Construct a 90% confidence interval for the true proportion.

$$x = 178, \quad n = 200, \quad 90\% \text{ CI}$$

Answer: ________________________________

17. In a random sample of 193 people, 156 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 156, \quad n = 193, \quad 99\% \text{ CI}$$

Answer: ________________________________

18. In a random sample of 163 people, 151 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 151, \quad n = 163, \quad 99\% \text{ CI}$$

Answer: ________________________________

19. In a random sample of 159 people, 49 had a particular trait. Construct a 95% confidence interval for the true proportion.

$$x = 49, \quad n = 159, \quad 95\% \text{ CI}$$

Answer: ________________________________

20. In a random sample of 250 people, 60 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 60, \quad n = 250, \quad 99\% \text{ CI}$$

Answer: ____

Confidence Interval for a Mean (t-distribution)

21. A sample of n = 25 gives x-bar = 86 and s = 19. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 86, \quad s = 19, \quad n = 25, \quad 95\% \text{ CI}$$

Answer: ____

22. A sample of n = 20 gives x-bar = 82 and s = 5. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 82, \quad s = 5, \quad n = 20, \quad 95\% \text{ CI}$$

Answer: ____

23. A sample of n = 15 gives x-bar = 91 and s = 11. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 91, \quad s = 11, \quad n = 15, \quad 95\% \text{ CI}$$

Answer: ____

24. A sample of n = 20 gives x-bar = 99 and s = 9. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 99, \quad s = 9, \quad n = 20, \quad 95\% \text{ CI}$$

Answer: ____

25. A sample of n = 5 gives x-bar = 114 and s = 20. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 114, \quad s = 20, \quad n = 5, \quad 95\% \text{ CI}$$

Answer: ____

26. A sample of n = 30 gives x-bar = 74 and s = 5. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 74, \quad s = 5, \quad n = 30, \quad 95\% \text{ CI}$$

Answer: ____

27. A sample of n = 5 gives x-bar = 119 and s = 8. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 119, \quad s = 8, \quad n = 5, \quad 95\%\ \text{CI}$$

Answer: ____

28. A sample of n = 15 gives x-bar = 86 and s = 12. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 86, \quad s = 12, \quad n = 15, \quad 95\%\ \text{CI}$$

Answer: ____

29. A sample of n = 30 gives x-bar = 107 and s = 17. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 107, \quad s = 17, \quad n = 30, \quad 95\%\ \text{CI}$$

Answer: ____

30. A sample of n = 25 gives x-bar = 80 and s = 19. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 80, \quad s = 19, \quad n = 25, \quad 95\%\ \text{CI}$$

Answer: ____

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ANSWER KEY & SOLUTIONS

Topics: Confidence Interval for a Mean (t-distribution), Confidence Interval for a Population Proportion, Confidence interval for a mean (σ known). All answers verified by independent computation.

Solutions

Confidence interval for a mean (σ known)

1. A sample of n = 16 has a sample mean of 131. The population standard deviation is sigma = 10. Construct the 99% confidence interval for the population mean mu.

$$\bar{x} = 131, \quad \sigma = 10, \quad n = 16, \quad 99\%\ \mathrm{CI}$$

→ *Critical value z* for a 99% interval = 2.576.*
→ *Standard error = sigma / sqrt(n) = 10/4 = 2.5.*
→ *Margin of error E = z* x SE = 2.576 x 2.5 = 6.44.*
→ *CI = sample mean +/- E = 131 +/- 6.44 = (124.56, 137.44).*

Answer: $$E = z^* \frac{\sigma}{\sqrt{n}} = 2.576 \cdot \frac{10}{4} = 6.44 \Rightarrow (124.56,\ 137.44)$$

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2. A sample of n = 100 has a sample mean of 84. The population standard deviation is sigma = 20. Construct the 95% confidence interval for the population mean mu.

$$\bar{x} = 84, \quad \sigma = 20, \quad n = 100, \quad 95\%\ \mathrm{CI}$$

→ *Critical value z* for a 95% interval = 1.96.*
→ *Standard error = sigma / sqrt(n) = 20/10 = 2.0.*
→ *Margin of error E = z* x SE = 1.96 x 2.0 = 3.92.*
→ *CI = sample mean +/- E = 84 +/- 3.92 = (80.08, 87.92).*

Answer: $$E = z^* \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{20}{10} = 3.92 \Rightarrow (80.08,\ 87.92)$$

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3. A sample of n = 25 has a sample mean of 120. The population standard deviation is sigma = 15. Construct the 99% confidence interval for the population mean mu.

$$\bar{x} = 120, \quad \sigma = 15, \quad n = 25, \quad 99\%\ \mathrm{CI}$$

→ *Critical value z* for a 99% interval = 2.576.*
→ *Standard error = sigma / sqrt(n) = 15/5 = 3.0.*
→ *Margin of error E = z* x SE = 2.576 x 3.0 = 7.728.*
→ *CI = sample mean +/- E = 120 +/- 7.728 = (112.272, 127.728).*

Answer: $$E = z^* \frac{\sigma}{\sqrt{n}} = 2.576 \cdot \frac{15}{5} = 7.728 \Rightarrow (112.272,\ 127.728)$$

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4. A sample of n = 25 has a sample mean of 81. The population standard deviation is sigma = 12. Construct the 90% confidence interval for the population mean mu.

$$\bar{x} = 81, \quad \sigma = 12, \quad n = 25, \quad 90\%\ \mathrm{CI}$$

→ *Critical value z* for a 90% interval = 1.645.*
→ *Standard error = sigma / sqrt(n) = 12/5 = 2.4.*
→ *Margin of error E = z* x SE = 1.645 x 2.4 = 3.948.*
→ *CI = sample mean +/- E = 81 +/- 3.948 = (77.052, 84.948).*

Answer: $$E = z^* \frac{\sigma}{\sqrt{n}} = 1.645 \cdot \frac{12}{5} = 3.948 \Rightarrow (77.052,\ 84.948)$$

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5. A sample of n = 36 has a sample mean of 67. The population standard deviation is sigma = 20. Construct the 95% confidence interval for the population mean mu.

$$\bar{x} = 67, \quad \sigma = 20, \quad n = 36, \quad 95\%\ \mathrm{CI}$$

→ *Critical value z* for a 95% interval = 1.96.*
→ *Standard error = sigma / sqrt(n) = 20/6 = 3.3333.*
→ *Margin of error E = z* x SE = 1.96 x 3.3333 = 6.533.*
→ *CI = sample mean +/- E = 67 +/- 6.533 = (60.467, 73.533).*

Answer: $$E = z^* \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{20}{6} = 6.533 \Rightarrow (60.467,\ 73.533)$$

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6. A sample of n = 16 has a sample mean of 55. The population standard deviation is sigma = 15. Construct the 90% confidence interval for the population mean mu.

$$\bar{x} = 55, \quad \sigma = 15, \quad n = 16, \quad 90\%\ \mathrm{CI}$$

→ *Critical value z* for a 90% interval = 1.645.*
→ *Standard error = sigma / sqrt(n) = 15/4 = 3.75.*
→ *Margin of error E = z* x SE = 1.645 x 3.75 = 6.169.*
→ *CI = sample mean +/- E = 55 +/- 6.169 = (48.831, 61.169).*

Answer: $$E = z^* \frac{\sigma}{\sqrt{n}} = 1.645 \cdot \frac{15}{4} = 6.169 \Rightarrow (48.831,\ 61.169)$$

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7. A sample of n = 25 has a sample mean of 89. The population standard deviation is sigma = 15. Construct the 95% confidence interval for the population mean mu.

$$\bar{x} = 89, \quad \sigma = 15, \quad n = 25, \quad 95\%\ \mathrm{CI}$$

→ *Critical value z* for a 95% interval = 1.96.*
→ *Standard error = sigma / sqrt(n) = 15/5 = 3.0.*
→ *Margin of error E = z* x SE = 1.96 x 3.0 = 5.88.*
→ *CI = sample mean +/- E = 89 +/- 5.88 = (83.12, 94.88).*

Answer: $$E = z^* \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{15}{5} = 5.88 \Rightarrow (83.12,\ 94.88)$$

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8. A sample of n = 36 has a sample mean of 106. The population standard deviation is sigma = 20. Construct the 95% confidence interval for the population mean mu.

$$\bar{x} = 106, \quad \sigma = 20, \quad n = 36, \quad 95\%\ \mathrm{CI}$$

→ *Critical value z* for a 95% interval = 1.96.*
→ *Standard error = sigma / sqrt(n) = 20/6 = 3.3333.*
→ *Margin of error E = z* x SE = 1.96 x 3.3333 = 6.533.*
→ *CI = sample mean +/- E = 106 +/- 6.533 = (99.467, 112.533).*

Answer: $$E = z^* \frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{20}{6} = 6.533 \Rightarrow (99.467,\ 112.533)$$

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9. A sample of n = 100 has a sample mean of 59. The population standard deviation is sigma = 15. Construct the 90% confidence interval for the population mean mu.

$$\bar{x} = 59, \quad \sigma = 15, \quad n = 100, \quad 90\%\ \mathrm{CI}$$

→ *Critical value z* for a 90% interval = 1.645.*
→ *Standard error = sigma / sqrt(n) = 15/10 = 1.5.*
→ *Margin of error E = z* x SE = 1.645 x 1.5 = 2.468.*
→ *CI = sample mean +/- E = 59 +/- 2.468 = (56.532, 61.468).*

Answer: $E = z^* \dfrac{\sigma}{\sqrt{n}} = 1.645 \cdot \dfrac{15}{10} = 2.468 \Rightarrow (56.532,\ 61.468)$

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10. A sample of n = 16 has a sample mean of 137. The population standard deviation is sigma = 10. Construct the 90% confidence interval for the population mean mu.

$$\bar{x} = 137, \quad \sigma = 10, \quad n = 16, \quad 90\%\ \mathrm{CI}$$

→ *Critical value z* for a 90% interval = 1.645.*
→ *Standard error = sigma / sqrt(n) = 10/4 = 2.5.*
→ *Margin of error E = z* x SE = 1.645 x 2.5 = 4.112.*
→ *CI = sample mean +/- E = 137 +/- 4.112 = (132.887, 141.113).*

Answer: $E = z^* \dfrac{\sigma}{\sqrt{n}} = 1.645 \cdot \dfrac{10}{4} = 4.112 \Rightarrow (132.887,\ 141.113)$

Confidence Interval for a Population Proportion

11. In a random sample of 246 people, 39 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 39, \quad n = 246, \quad 99\% \text{ CI}$$

→ *Sample proportion: p-hat = x/n = 39/246 = 0.1585.*
→ *Critical value: z* = 2.576 for 99% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.0233.*
→ *Margin of error: E = z* * SE = 2.576 * 0.0233 = 0.06.*
→ *99% CI: (0.1585 - 0.06, 0.1585 + 0.06) = (0.0985, 0.2185).*

Answer: $\hat{p} = 0.1585,\ E = 0.06 \Rightarrow (0.0985,\ 0.2185)$

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12. In a random sample of 304 people, 49 had a particular trait. Construct a 90% confidence interval for the true proportion.

$$x = 49, \quad n = 304, \quad 90\% \text{ CI}$$

→ *Sample proportion: p-hat = x/n = 49/304 = 0.1612.*
→ *Critical value: z* = 1.645 for 90% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.0211.*
→ *Margin of error: E = z* * SE = 1.645 * 0.0211 = 0.0347.*
→ *90% CI: (0.1612 - 0.0347, 0.1612 + 0.0347) = (0.1265, 0.1959).*

Answer: $\hat{p} = 0.1612,\ E = 0.0347 \Rightarrow (0.1265,\ 0.1959)$

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13. In a random sample of 276 people, 47 had a particular trait. Construct a 95% confidence interval for the true proportion.

$$x = 47, \quad n = 276, \quad 95\% \text{ CI}$$

→ *Sample proportion: p-hat = x/n = 47/276 = 0.1703.*
→ *Critical value: z* = 1.96 for 95% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.0226.*
→ *Margin of error: E = z* * SE = 1.96 * 0.0226 = 0.0443.*
→ *95% CI: (0.1703 - 0.0443, 0.1703 + 0.0443) = (0.126, 0.2146).*

Answer: $\hat{p} = 0.1703,\ E = 0.0443 \Rightarrow (0.126,\ 0.2146)$

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14. In a random sample of 127 people, 98 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 98, \quad n = 127, \quad 99\% \text{ CI}$$

→ *Sample proportion: p-hat = x/n = 98/127 = 0.7717.*
→ *Critical value: z* = 2.576 for 99% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.0372.*
→ *Margin of error: E = z* * SE = 2.576 * 0.0372 = 0.0959.*
→ *99% CI: (0.7717 - 0.0959, 0.7717 + 0.0959) = (0.6758, 0.8676).*

Answer: $\hat{p} = 0.7717,\ E = 0.0959 \Rightarrow (0.6758,\ 0.8676)$

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15. In a random sample of 200 people, 53 had a particular trait. Construct a 90% confidence interval for the true proportion.

$$x = 53, \quad n = 200, \quad 90\%\ \text{CI}$$

→ *Sample proportion: p-hat = x/n = 53/200 = 0.265.*
→ *Critical value: z* = 1.645 for 90% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.0312.*
→ *Margin of error: E = z* * SE = 1.645 * 0.0312 = 0.0513.*
→ *90% CI: (0.265 - 0.0513, 0.265 + 0.0513) = (0.2137, 0.3163).*

Answer: $\hat{p} = 0.265,\ E = 0.0513 \Rightarrow (0.2137,\ 0.3163)$

16. In a random sample of 200 people, 178 had a particular trait. Construct a 90% confidence interval for the true proportion.

$$x = 178, \quad n = 200, \quad 90\%\ \text{CI}$$

→ *Sample proportion: p-hat = x/n = 178/200 = 0.89.*
→ *Critical value: z* = 1.645 for 90% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.0221.*
→ *Margin of error: E = z* * SE = 1.645 * 0.0221 = 0.0364.*
→ *90% CI: (0.89 - 0.0364, 0.89 + 0.0364) = (0.8536, 0.9264).*

Answer: $\hat{p} = 0.89,\ E = 0.0364 \Rightarrow (0.8536,\ 0.9264)$

17. In a random sample of 193 people, 156 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 156, \quad n = 193, \quad 99\%\ \text{CI}$$

→ *Sample proportion: p-hat = x/n = 156/193 = 0.8083.*
→ *Critical value: z* = 2.576 for 99% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.0283.*
→ *Margin of error: E = z* * SE = 2.576 * 0.0283 = 0.073.*
→ *99% CI: (0.8083 - 0.073, 0.8083 + 0.073) = (0.7353, 0.8813).*

Answer: $\hat{p} = 0.8083,\ E = 0.073 \Rightarrow (0.7353,\ 0.8813)$

18. In a random sample of 163 people, 151 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 151, \quad n = 163, \quad 99\%\ \text{CI}$$

→ *Sample proportion: p-hat = x/n = 151/163 = 0.9264.*
→ *Critical value: z* = 2.576 for 99% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.0205.*
→ *Margin of error: E = z* * SE = 2.576 * 0.0205 = 0.0527.*
→ *99% CI: (0.9264 - 0.0527, 0.9264 + 0.0527) = (0.8737, 0.9791).*

Answer: $\hat{p} = 0.9264,\ E = 0.0527 \Rightarrow (0.8737,\ 0.9791)$

19. In a random sample of 159 people, 49 had a particular trait. Construct a 95% confidence interval for the true proportion.

$$x = 49, \quad n = 159, \quad 95\%\ \text{CI}$$

→ *Sample proportion: p-hat = x/n = 49/159 = 0.3082.*
→ *Critical value: z* = 1.96 for 95% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.0366.*
→ *Margin of error: E = z* * SE = 1.96 * 0.0366 = 0.0718.*
→ *95% CI: (0.3082 - 0.0718, 0.3082 + 0.0718) = (0.2364, 0.38).*

Answer: $\hat{p} = 0.3082,\ E = 0.0718 \Rightarrow (0.2364,\ 0.38)$

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20. In a random sample of 250 people, 60 had a particular trait. Construct a 99% confidence interval for the true proportion.

$$x = 60, \quad n = 250, \quad 99\%\ \text{CI}$$

→ *Sample proportion: p-hat = x/n = 60/250 = 0.24.*
→ *Critical value: z* = 2.576 for 99% CI.*
→ *Standard error: SE = sqrt(p-hat*(1-p-hat)/n) = 0.027.*
→ *Margin of error: E = z* * SE = 2.576 * 0.027 = 0.0696.*
→ *99% CI: (0.24 - 0.0696, 0.24 + 0.0696) = (0.1704, 0.3096).*

Answer: $\hat{p} = 0.24,\ E = 0.0696 \Rightarrow (0.1704,\ 0.3096)$

Confidence Interval for a Mean (t-distribution)

21. A sample of n = 25 gives x-bar = 86 and s = 19. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 86, \quad s = 19, \quad n = 25, \quad 95\%\ \mathrm{CI}$$

→ *Degrees of freedom: df = n - 1 = 25 - 1 = 24.*
→ *Critical value: t* = 2.064 (95% CI, df=24).*
→ *Standard error: SE = s / sqrt(n) = 19 / sqrt(25) = 3.8.*
→ *Margin of error: E = t* * SE = 2.064 * 3.8 = 7.843.*
→ *95% CI: (86 - 7.843, 86 + 7.843) = (78.157, 93.843).*

Answer: $t^* = 2.064,\ SE = 3.8,\ E = 7.843 \Rightarrow (78.157,\ 93.843)$

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22. A sample of n = 20 gives x-bar = 82 and s = 5. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 82, \quad s = 5, \quad n = 20, \quad 95\%\ \mathrm{CI}$$

→ *Degrees of freedom: df = n - 1 = 20 - 1 = 19.*
→ *Critical value: t* = 2.093 (95% CI, df=19).*
→ *Standard error: SE = s / sqrt(n) = 5 / sqrt(20) = 1.118.*
→ *Margin of error: E = t* * SE = 2.093 * 1.118 = 2.34.*
→ *95% CI: (82 - 2.34, 82 + 2.34) = (79.66, 84.34).*

Answer: $t^* = 2.093,\ SE = 1.118,\ E = 2.34 \Rightarrow (79.66,\ 84.34)$

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23. A sample of n = 15 gives x-bar = 91 and s = 11. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 91, \quad s = 11, \quad n = 15, \quad 95\%\ \mathrm{CI}$$

→ *Degrees of freedom: df = n - 1 = 15 - 1 = 14.*
→ *Critical value: t* = 2.145 (95% CI, df=14).*
→ *Standard error: SE = s / sqrt(n) = 11 / sqrt(15) = 2.8402.*
→ *Margin of error: E = t* * SE = 2.145 * 2.8402 = 6.092.*
→ *95% CI: (91 - 6.092, 91 + 6.092) = (84.908, 97.092).*

Answer: $t^* = 2.145,\ SE = 2.8402,\ E = 6.092 \Rightarrow (84.908,\ 97.092)$

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24. A sample of n = 20 gives x-bar = 99 and s = 9. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 99, \quad s = 9, \quad n = 20, \quad 95\%\ \mathrm{CI}$$

→ *Degrees of freedom: df = n - 1 = 20 - 1 = 19.*
→ *Critical value: t* = 2.093 (95% CI, df=19).*
→ *Standard error: SE = s / sqrt(n) = 9 / sqrt(20) = 2.0125.*
→ *Margin of error: E = t* * SE = 2.093 * 2.0125 = 4.212.*
→ *95% CI: (99 - 4.212, 99 + 4.212) = (94.788, 103.212).*

Answer: $t^* = 2.093,\ SE = 2.0125,\ E = 4.212 \Rightarrow (94.788,\ 103.212)$

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25. A sample of n = 5 gives x-bar = 114 and s = 20. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 114, \quad s = 20, \quad n = 5, \quad 95\%\ \text{CI}$$

→ *Degrees of freedom: df = n - 1 = 5 - 1 = 4.*
→ *Critical value: t* = 2.776 (95% CI, df=4).*
→ *Standard error: SE = s / sqrt(n) = 20 / sqrt(5) = 8.9443.*
→ *Margin of error: E = t* * SE = 2.776 * 8.9443 = 24.829.*
→ *95% CI: (114 - 24.829, 114 + 24.829) = (89.171, 138.829).*

Answer: $t^* = 2.776,\ SE = 8.9443,\ E = 24.829\ \Rightarrow\ (89.171,\ 138.829)$

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26. A sample of n = 30 gives x-bar = 74 and s = 5. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 74, \quad s = 5, \quad n = 30, \quad 95\%\ \text{CI}$$

→ *Degrees of freedom: df = n - 1 = 30 - 1 = 29.*
→ *Critical value: t* = 2.045 (95% CI, df=29).*
→ *Standard error: SE = s / sqrt(n) = 5 / sqrt(30) = 0.9129.*
→ *Margin of error: E = t* * SE = 2.045 * 0.9129 = 1.867.*
→ *95% CI: (74 - 1.867, 74 + 1.867) = (72.133, 75.867).*

Answer: $t^* = 2.045,\ SE = 0.9129,\ E = 1.867\ \Rightarrow\ (72.133,\ 75.867)$

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27. A sample of n = 5 gives x-bar = 119 and s = 8. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 119, \quad s = 8, \quad n = 5, \quad 95\%\ \text{CI}$$

→ *Degrees of freedom: df = n - 1 = 5 - 1 = 4.*
→ *Critical value: t* = 2.776 (95% CI, df=4).*
→ *Standard error: SE = s / sqrt(n) = 8 / sqrt(5) = 3.5777.*
→ *Margin of error: E = t* * SE = 2.776 * 3.5777 = 9.932.*
→ *95% CI: (119 - 9.932, 119 + 9.932) = (109.068, 128.932).*

Answer: $t^* = 2.776,\ SE = 3.5777,\ E = 9.932\ \Rightarrow\ (109.068,\ 128.932)$

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28. A sample of n = 15 gives x-bar = 86 and s = 12. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 86, \quad s = 12, \quad n = 15, \quad 95\%\ \text{CI}$$

→ *Degrees of freedom: df = n - 1 = 15 - 1 = 14.*
→ *Critical value: t* = 2.145 (95% CI, df=14).*
→ *Standard error: SE = s / sqrt(n) = 12 / sqrt(15) = 3.0984.*
→ *Margin of error: E = t* * SE = 2.145 * 3.0984 = 6.646.*
→ *95% CI: (86 - 6.646, 86 + 6.646) = (79.354, 92.646).*

Answer: $t^* = 2.145,\ SE = 3.0984,\ E = 6.646\ \Rightarrow\ (79.354,\ 92.646)$

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29. A sample of n = 30 gives x-bar = 107 and s = 17. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 107, \quad s = 17, \quad n = 30, \quad 95\%\ \text{CI}$$

→ *Degrees of freedom: df = n - 1 = 30 - 1 = 29.*
→ *Critical value: t* = 2.045 (95% CI, df=29).*
→ *Standard error: SE = s / sqrt(n) = 17 / sqrt(30) = 3.1038.*
→ *Margin of error: E = t* * SE = 2.045 * 3.1038 = 6.347.*
→ *95% CI: (107 - 6.347, 107 + 6.347) = (100.653, 113.347).*

Answer: $t^* = 2.045,\ SE = 3.1038,\ E = 6.347 \Rightarrow (100.653,\ 113.347)$

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30. A sample of n = 25 gives x-bar = 80 and s = 19. Build a 95% confidence interval for the population mean using the t-distribution.

$$\bar{x} = 80, \quad s = 19, \quad n = 25, \quad 95\%\ \text{CI}$$

→ *Degrees of freedom: df = n - 1 = 25 - 1 = 24.*
→ *Critical value: t* = 2.064 (95% CI, df=24).*
→ *Standard error: SE = s / sqrt(n) = 19 / sqrt(25) = 3.8.*
→ *Margin of error: E = t* * SE = 2.064 * 3.8 = 7.843.*
→ *95% CI: (80 - 7.843, 80 + 7.843) = (72.157, 87.843).*

Answer: $t^* = 2.064,\ SE = 3.8,\ E = 7.843 \Rightarrow (72.157,\ 87.843)$
